

Hazard_comparison.R

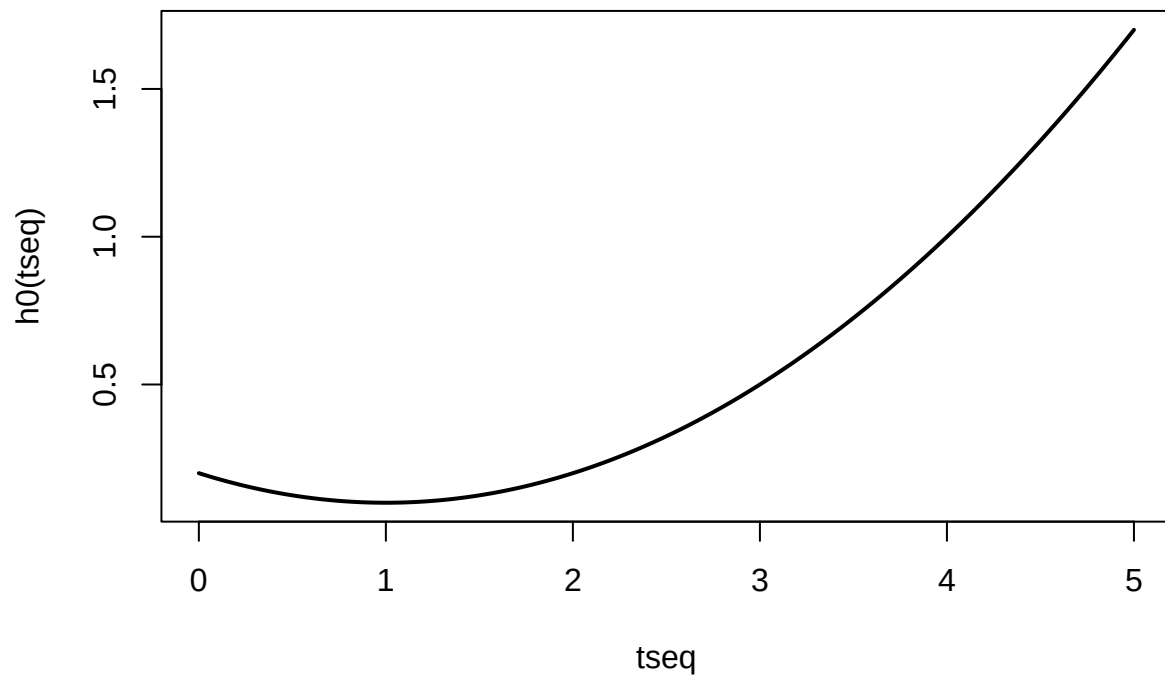
hputter

2025-11-14

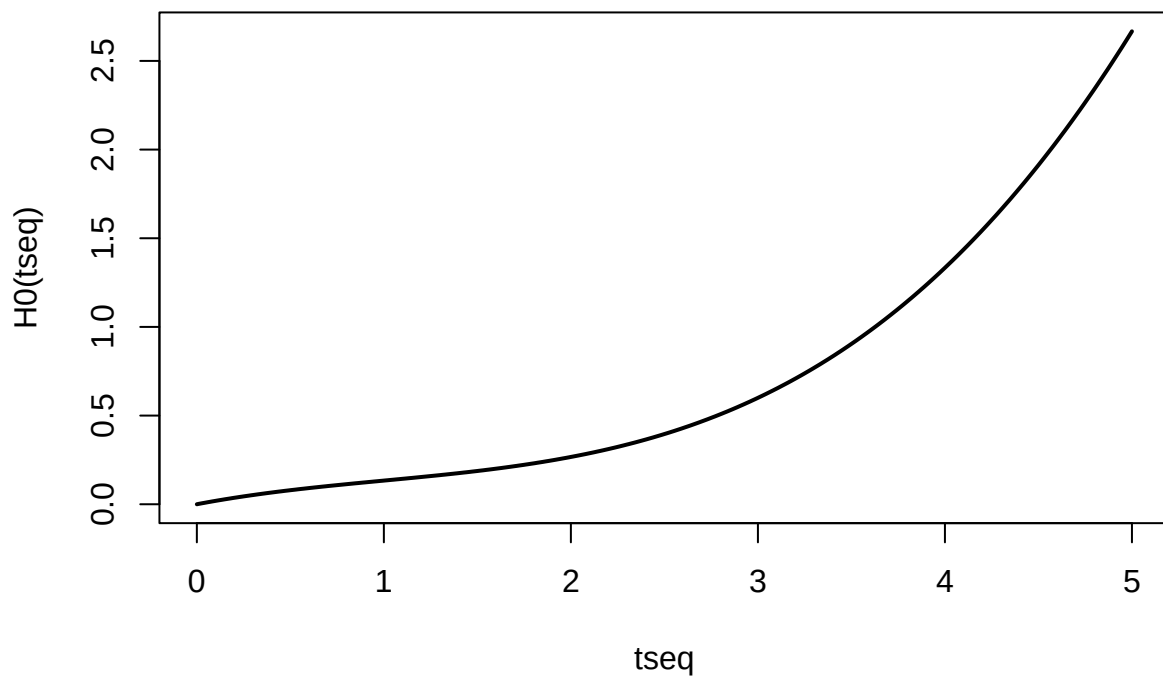
```
tseq <- seq(0, 5, by = 0.01)

h0 <- function(t) 0.1 * (t - 1)^2 + 0.1
H0 <- function(t) 0.1/3 * (t - 1)^3 + 0.1 * t + 0.1/3
S0 <- function(t) exp(-H0(t))

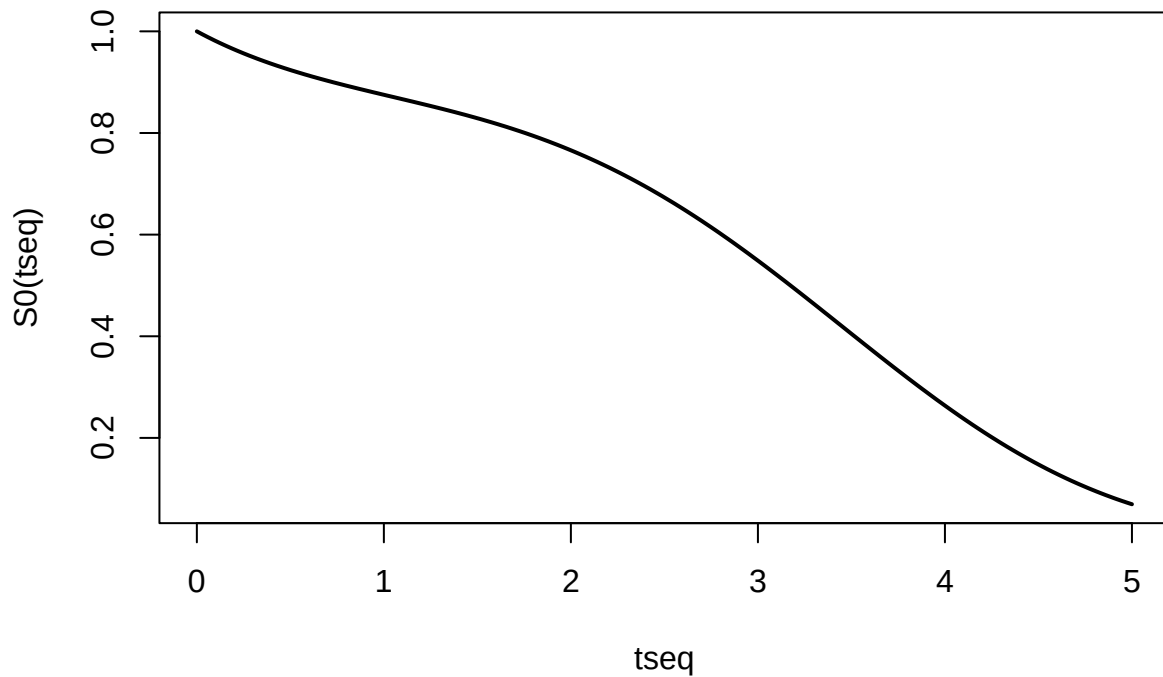
plot(tseq, h0(tseq), type = "l", lwd = 2)
```



```
plot(tseq, H0(tseq), type = "l", lwd = 2)
```



```
plot(tseq, S0(tseq), type = "l", lwd = 2)
```



```

RMST5 <- integrate(S0, lower = 0, upper = 5)
RMST5

## 2.981754 with absolute error < 2.4e-13

RMST5 <- RMST5$value
## Find beta such that RMST5 for X=1 decreases with 1
# Cox model
f <- function(bet) {
  S1_cox <- function(t) exp(-H0(t) * exp(bet))
  integrate(S1_cox, lower = 0, upper = 5)$value - RMST5 + 1
}
ur <- uniroot(f, lower = 0, upper = 1)
ur

## $root
## [1] 0.886777
##
## $f.root
## [1] 1.996898e-05
##
## $iter
## [1] 3
##
## $init.it
## [1] NA
##

```

```

## $estim.prec
## [1] 6.103516e-05

bet_cox <- ur$root
hplus_cox <- function(t) h0(t) * exp(bet_cox)
hminus_cox <- function(t) h0(t) * exp(-bet_cox)
Splus_cox <- function(t) S0(t) ^ exp(bet_cox)
Sminus_cox <- function(t) S0(t) ^ exp(-bet_cox)
Hplus_cox <- function(t) H0(t) * exp(bet_cox)
Hminus_cox <- function(t) H0(t) * exp(-bet_cox)
# Additive hazards model
f <- function(bet) {
  S1_ah <- function(t) exp(-H0(t) - bet * t)
  integrate(S1_ah, lower = 0, upper = 5)$value - RMST5 + 1
}
ur <- uniroot(f, lower = 0, upper = 1)
ur

## $root
## [1] 0.2489588
##
## $f.root
## [1] -7.149806e-06
##
## $iter
## [1] 6
##
## $init.it
## [1] NA
##
## $estim.prec
## [1] 6.103516e-05

bet_ah <- ur$root
hplus_ah <- function(t) h0(t) + bet_ah
hminus_ah <- function(t) h0(t) - bet_ah
Splus_ah <- function(t) S0(t) * exp(-bet_ah * t)
Sminus_ah <- function(t) S0(t) * exp(bet_ah * t)
Hplus_ah <- function(t) H0(t) + bet_ah * t
Hminus_ah <- function(t) H0(t) - bet_ah * t
# AFT model
f <- function(bet) {
  S1_AFT <- function(t) S0(t * exp(-bet))
  integrate(S1_AFT, lower = 0, upper = 5)$value - RMST5 + 1
}
ur <- uniroot(f, lower = -1, upper = 0)
ur

## $root
## [1] -0.419737
##
## $f.root
## [1] -6.280649e-06
##
## $iter
## [1] 5

```

```

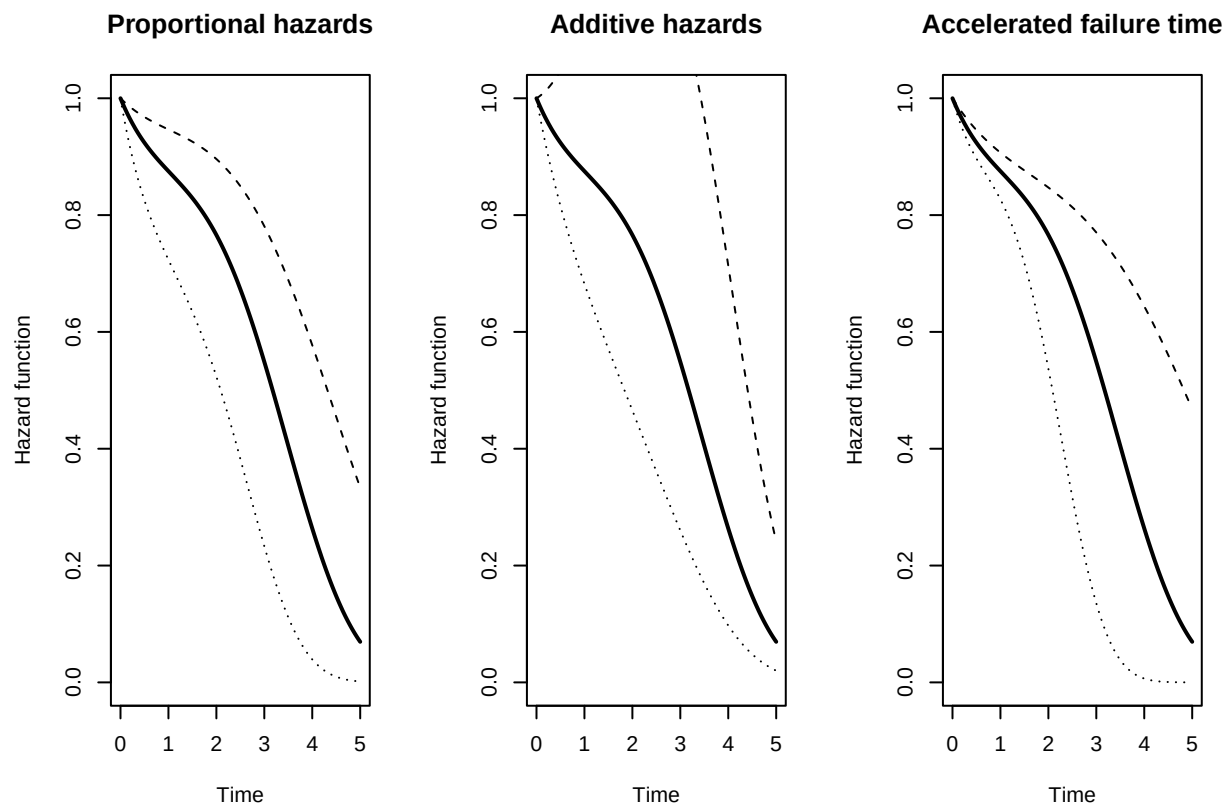
##
## $init.it
## [1] NA
##
## $estim.prec
## [1] 6.103516e-05

bet_AFT <- ur$root
hplus_AFT <- function(t) h0(t * exp(-bet_AFT)) * exp(-bet_AFT)
hminus_AFT <- function(t) h0(t * exp(bet_AFT)) * exp(bet_AFT)
Splus_AFT <- function(t) S0(t * exp(-bet_AFT))
Sminus_AFT <- function(t) S0(t * exp(bet_AFT))
Hplus_AFT <- function(t) H0(t * exp(-bet_AFT))
Hminus_AFT <- function(t) H0(t * exp(bet_AFT))

### Plot
tseq <- seq(0, 5, by = 0.01)

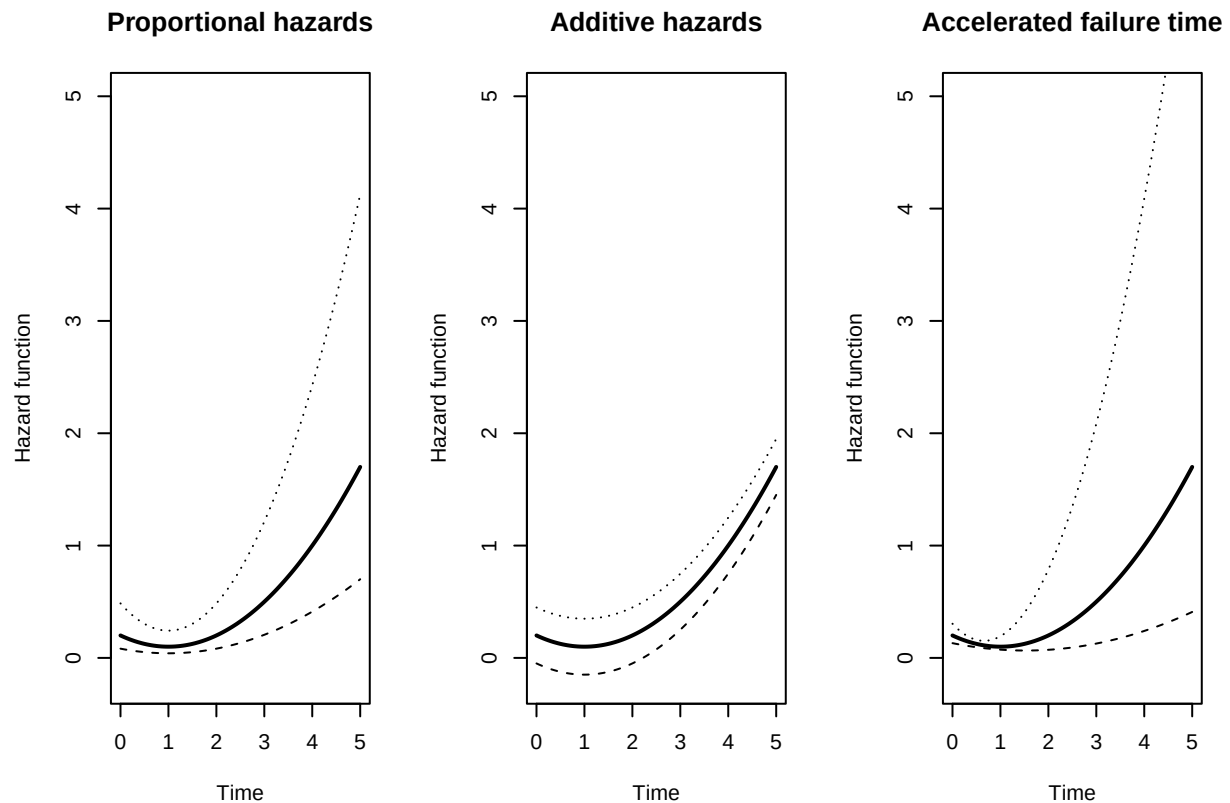
# pdf("Survival_curves.pdf", width = 8, height = 3)
## Survival curves
my_ylim = c(0, 1)
par(mfrow = c(1, 3))
# Cox
plot(tseq, S0(tseq), type = "l", lwd = 2, ylim = my_ylim,
      xlab = "Time", ylab = "Hazard function")
lines(tseq, Splus_cox(tseq), lty = 3)
lines(tseq, Sminus_cox(tseq), lty = 2)
title(main = "Proportional hazards")
# Additive hazards
plot(tseq, S0(tseq), type = "l", lwd = 2, ylim = my_ylim,
      xlab = "Time", ylab = "Hazard function")
lines(tseq, Splus_ah(tseq), lty = 3)
lines(tseq, Sminus_ah(tseq), lty = 2)
title(main = "Additive hazards")
# AFT
plot(tseq, S0(tseq), type = "l", lwd = 2, ylim = my_ylim,
      xlab = "Time", ylab = "Hazard function")
lines(tseq, Splus_AFT(tseq), lty = 3)
lines(tseq, Sminus_AFT(tseq), lty = 2)
title(main = "Accelerated failure time")

```



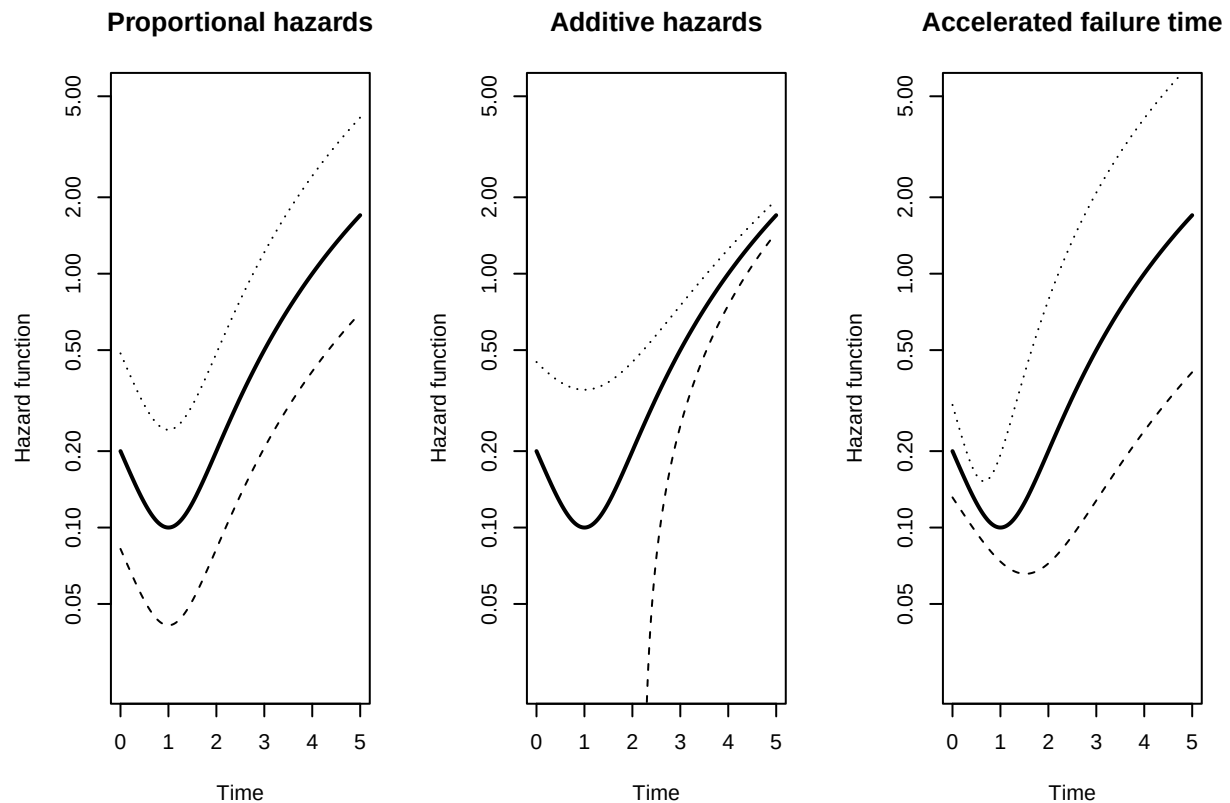
```
# dev.off()

# pdf("Hazard_curves.pdf", width = 8, height = 3)
## Hazards (original scale)
my_ylim = c(-0.2, 5)
par(mfrow = c(1, 3))
# Cox
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, hplus_cox(tseq), lty = 3)
lines(tseq, hminus_cox(tseq), lty = 2)
title(main = "Proportional hazards")
# Additive hazards
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, hplus_ah(tseq), lty = 3)
lines(tseq, hminus_ah(tseq), lty = 2)
title(main = "Additive hazards")
# AFT
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, hplus_AFT(tseq), lty = 3)
lines(tseq, hminus_AFT(tseq), lty = 2)
title(main = "Accelerated failure time")
```



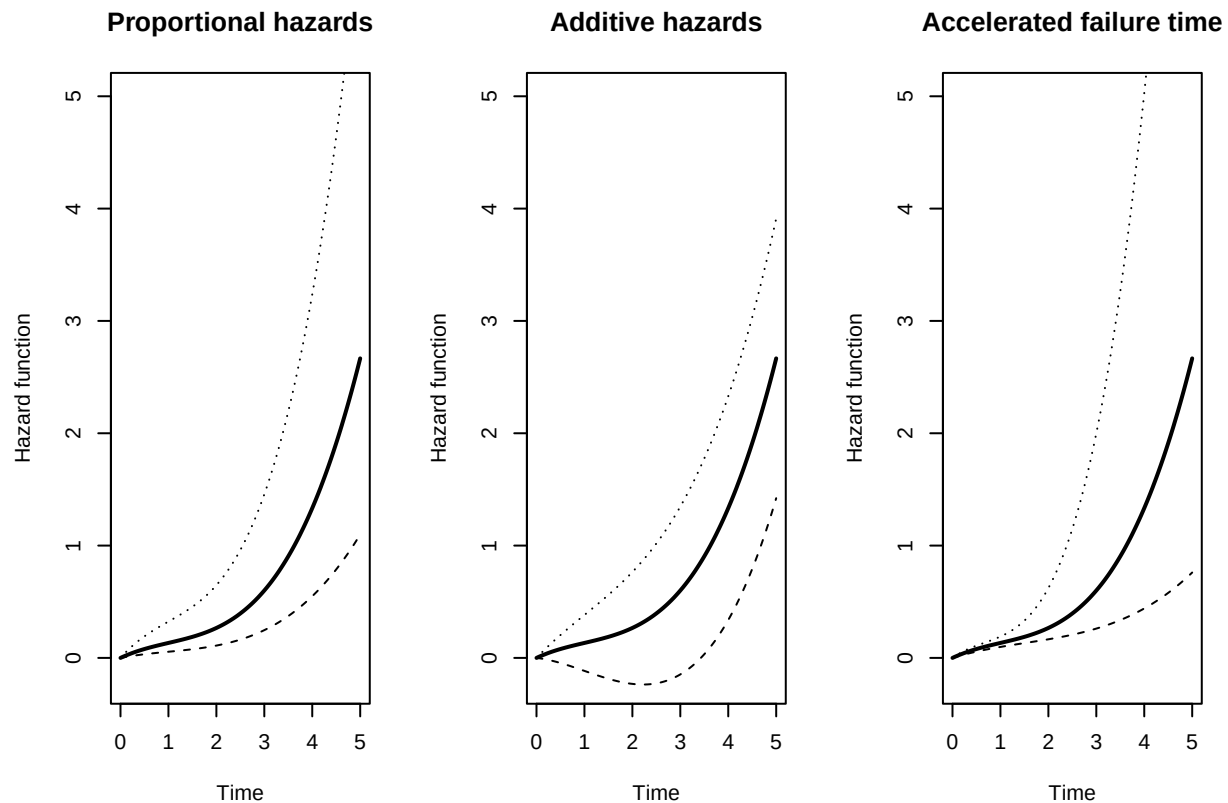
```
# dev.off()

# pdf("Hazard_curves_log.pdf", width = 8, height = 3)
## Hazards (log-scale y-axis)
my_ylim = c(0.025, 5)
par(mfrow = c(1, 3))
# Cox
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function", log = "y")
lines(tseq, hplus_cox(tseq), lty = 3)
lines(tseq, hminus_cox(tseq), lty = 2)
title(main = "Proportional hazards")
# Additive hazards
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function", log = "y")
lines(tseq, hplus_ah(tseq), lty = 3)
lines(tseq, hminus_ah(tseq), lty = 2)
title(main = "Additive hazards")
# AFT
plot(tseq, h0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function", log = "y")
lines(tseq, hplus_AFT(tseq), lty = 3)
lines(tseq, hminus_AFT(tseq), lty = 2)
title(main = "Accelerated failure time")
```



```
# dev.off()

# pdf("Cum_hazard_curves.pdf", width = 8, height = 3)
## Hazards (original scale)
my_ylim = c(-0.2, 5)
par(mfrow = c(1, 3))
# Cox
plot(tseq, H0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, Hplus_cox(tseq), lty = 3)
lines(tseq, Hminus_cox(tseq), lty = 2)
title(main = "Proportional hazards")
# Additive hazards
plot(tseq, H0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, Hplus_ah(tseq), lty = 3)
lines(tseq, Hminus_ah(tseq), lty = 2)
title(main = "Additive hazards")
# AFT
plot(tseq, H0(tseq), type = "l", lwd = 2, ylim = my_ylim,
     xlab = "Time", ylab = "Hazard function")
lines(tseq, Hplus_AFT(tseq), lty = 3)
lines(tseq, Hminus_AFT(tseq), lty = 2)
title(main = "Accelerated failure time")
```



```
# dev.off()
```

```
# When is the hazard for additive hazards model for  $X=-1$  negative?
```

```
hminus_ah(tseq[223:224])
```

```
## [1] -0.0001188201 0.0023311799
```

```
tseq[223:224]
```

```
## [1] 2.22 2.23
```

```
# When is the survival curve for additive hazards model for  $X=-1$  below one?
```

```
Sminus_ah(tseq[343:344])
```

```
## [1] 1.0036964 0.9992988
```

```
tseq[343:344]
```

```
## [1] 3.42 3.43
```

```
# Wat ik nog extra kan doen (misschien alleen voor github?) is het volgende.
# Ik kan de meer flexibele \ah en \aft modellen gebruiken, en ervoor zorgen
# dat ik in beide gevallen (\eqref{eq:ah} voor \ah, \eqref{eq:td_AFT} voor AFT)
#  $\beta(t)$  zodanig kies dat zowel voor  $X=0$  als  $X=1$  de hazards en dus
# survival curves exact gelijk zijn voor de drie modellen. Ik hoop dat ik dat
# voor elkaar krijg. Voor \ah kan dat makkelijk, voor AFT weet ik nog niet.
# Vervolgens laten zien hoe de (log) hazard en survival curves eruit zien
# voor  $X=-1$ . Heb ik nog niet gedaan, maar kan ook inzicht opleveren.
```